Decomposition analyses (AI; reordered; dredge-selected best models refit with REML)

30 April 2026

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0.1 Packages

0.2 Data (CSV, filtering, AI extraction)

Data summary

CSV_used	N	Sites	AI_min	AI_max
C:/Users/RTHOR95/OneDrive - The University of Liverpool/R/joel_termites/Termite_Decomp/Termite_Decomp/Updated_DB4_NEW.csv	35	10	0.353	1.792

0.3 Helpers (print-friendly tables, best-model refit, R², plotting style)

1 1) Termite absolute (Term_Absolute)

(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
57.41	-15.17	NA	NA	NA	NA	4	-139.36	298.29	0.00	0.28
45.32	NA	NA	NA	NA	NA	3	-144.03	299.30	1.01	0.17
73.95	-27.30	NA	+	NA	NA	5	-135.50	299.62	1.34	0.14

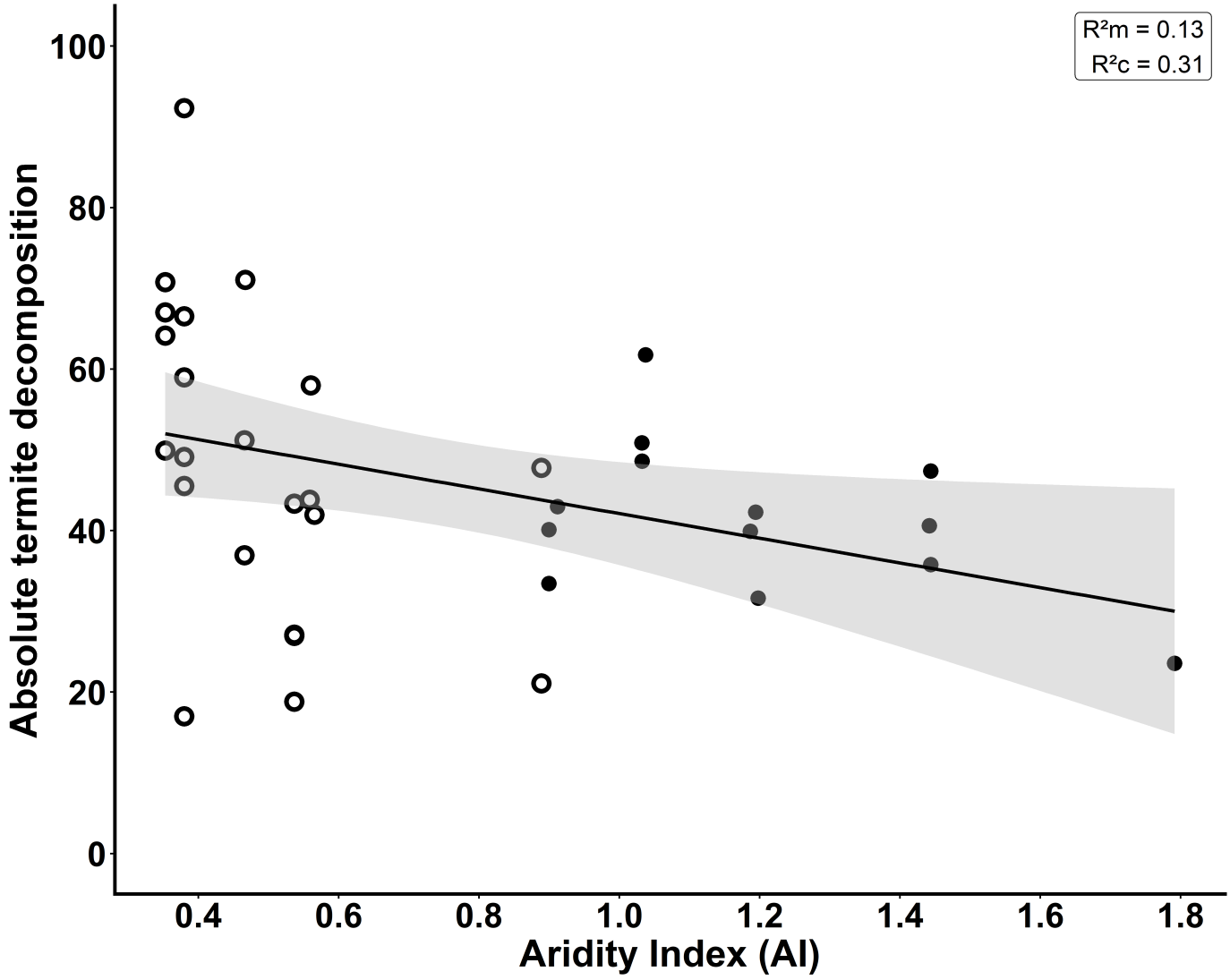
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
30.39	-17.66	1.16	NA	NA	NA	5	-137.72	300.55	2.26	0.09
9.03	-15.66	1.97	+	+	NA	7	-128.41	301.02	2.73	0.07
60.24	-16.14	NA	+	+	NA	6	-130.61	301.09	2.81	0.07
42.08	NA	NA	+	NA	NA	4	-140.87	301.13	2.85	0.07
44.99	NA	0.01	NA	NA	NA	4	-142.61	302.01	3.72	0.04
47.24	-29.30	1.13	+	NA	NA	6	-133.90	302.05	3.76	0.04
31.76	NA	0.41	+	NA	NA	5	-139.39	303.99	5.70	0.02
50.18	-15.93	0.39	+	+	+	8	-125.05	305.03	6.74	0.01
134.84	-30.42	-2.20	+	NA	+	7	-130.37	305.32	7.03	0.01
-172.04	NA	8.27	+	NA	+	6	-135.71	306.80	8.51	0.00

Selected_model_formula
Term_Absolute ~ AI + (1 | Site)

nobs	AIC	BIC	logLik	sigma
35	286.73	292.95	-139.36	14.11

term	estimate	std_error	df	t_value
(Intercept)	57.41	7.21	8.56	7.96
AI	-15.17	8.08	11.12	-1.88

R2m	R2c
0.13	0.31



2 2) Microbe absolute (Microbe_Absolute)

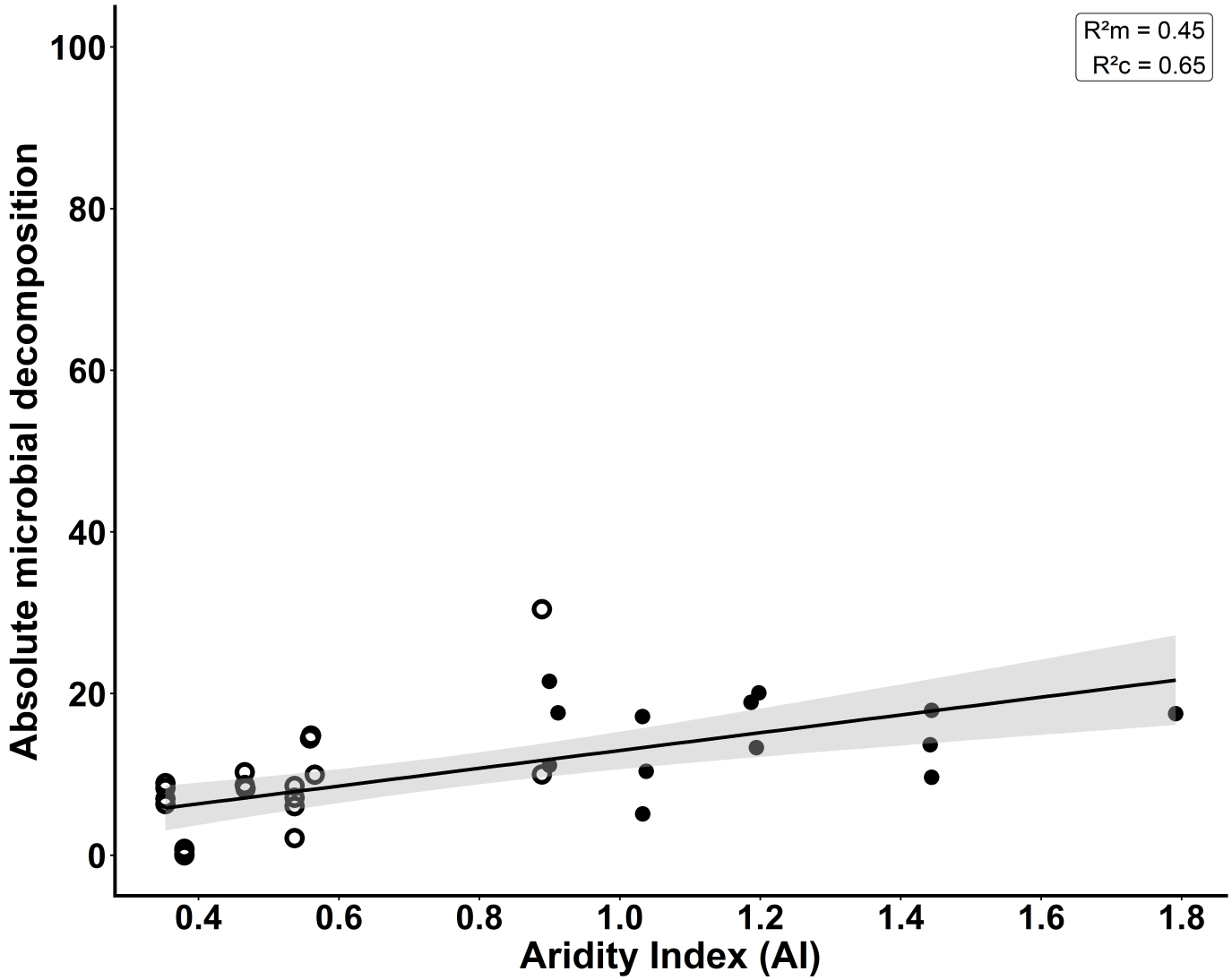
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
-23.00	7.15	1.13	NA	NA	NA	5	-100.21	219.27	0.00	0.30
3.53	9.12	NA	NA	NA	NA	4	-102.33	220.69	1.42	0.15
12.97	1.72	NA	+	+	NA	6	-94.49	220.91	1.64	0.13
-8.90	2.41	0.81	+	+	NA	7	-93.13	221.32	2.04	0.11
-26.37	9.48	1.13	+	NA	NA	6	-97.83	221.79	2.52	0.08
-28.29	NA	1.57	NA	NA	NA	4	-104.57	222.06	2.79	0.07
0.11	11.56	NA	+	NA	NA	5	-99.91	222.96	3.69	0.05
-22.72	NA	1.41	+	NA	NA	5	-102.29	223.37	4.10	0.04
11.13	NA	NA	NA	NA	NA	3	-107.44	224.74	5.47	0.02
1.45	2.24	0.42	+	+	+	8	-90.73	225.11	5.84	0.02
-53.40	10.06	2.15	+	NA	+	7	-95.34	225.14	5.87	0.02
60.22	NA	-1.80	+	NA	+	6	-99.62	226.06	6.79	0.01
12.83	NA	NA	+	NA	NA	4	-105.06	226.06	6.79	0.01

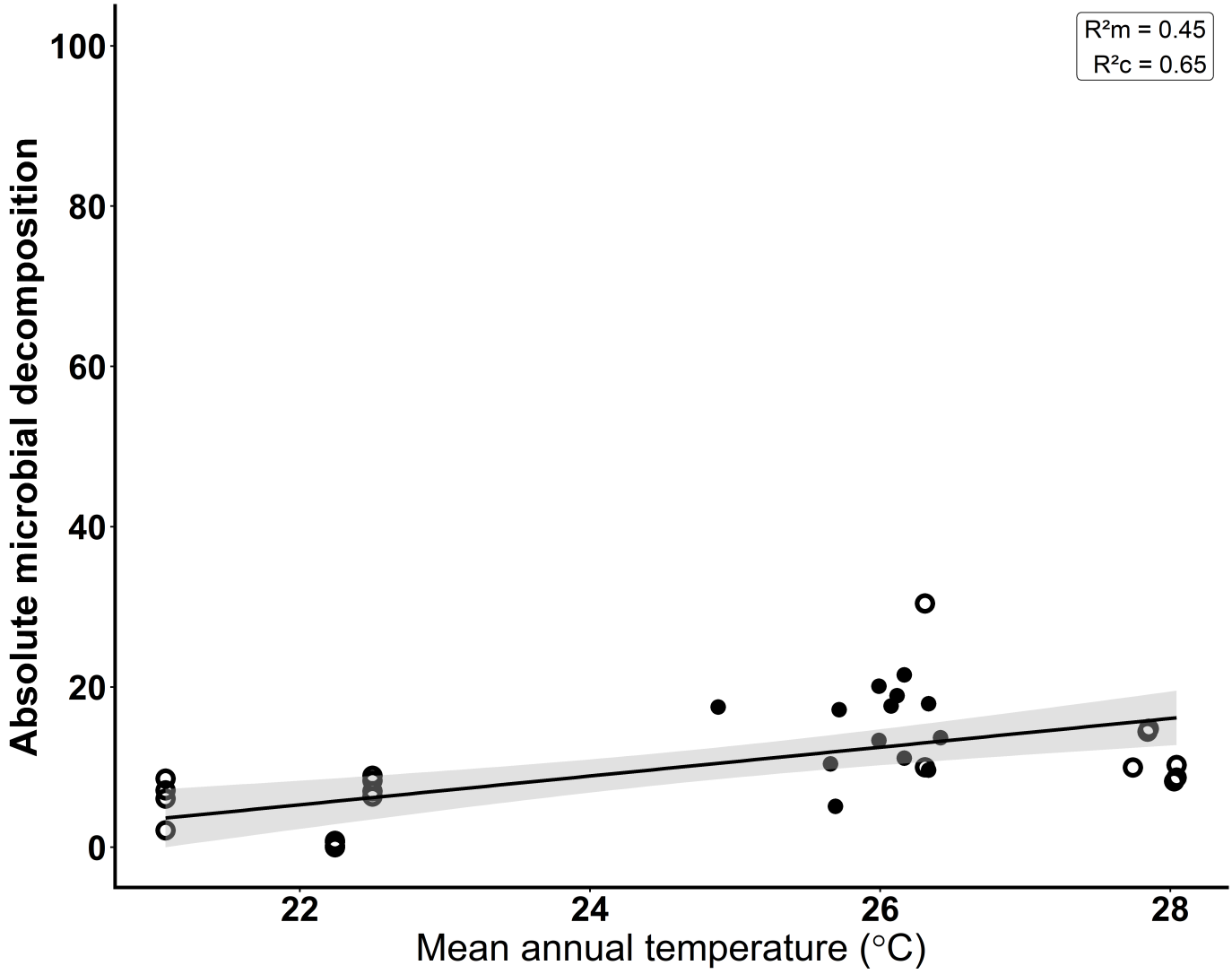
Selected_model_formula
Microbe_Absolute ~ AI + Avg_Temp + (1 | Site)

nobs	AIC	BIC	logLik	sigma
35	210.42	218.2	-100.21	4.21

term	estimate	std_error	df	t_value
(Intercept)	-23.00	13.51	7.58	-1.70
AI	7.15	3.08	11.20	2.32
Avg_Temp	1.13	0.57	8.09	1.99

R2m	R2c
0.45	0.64





3 3) Total decomposition (Term_Loss)

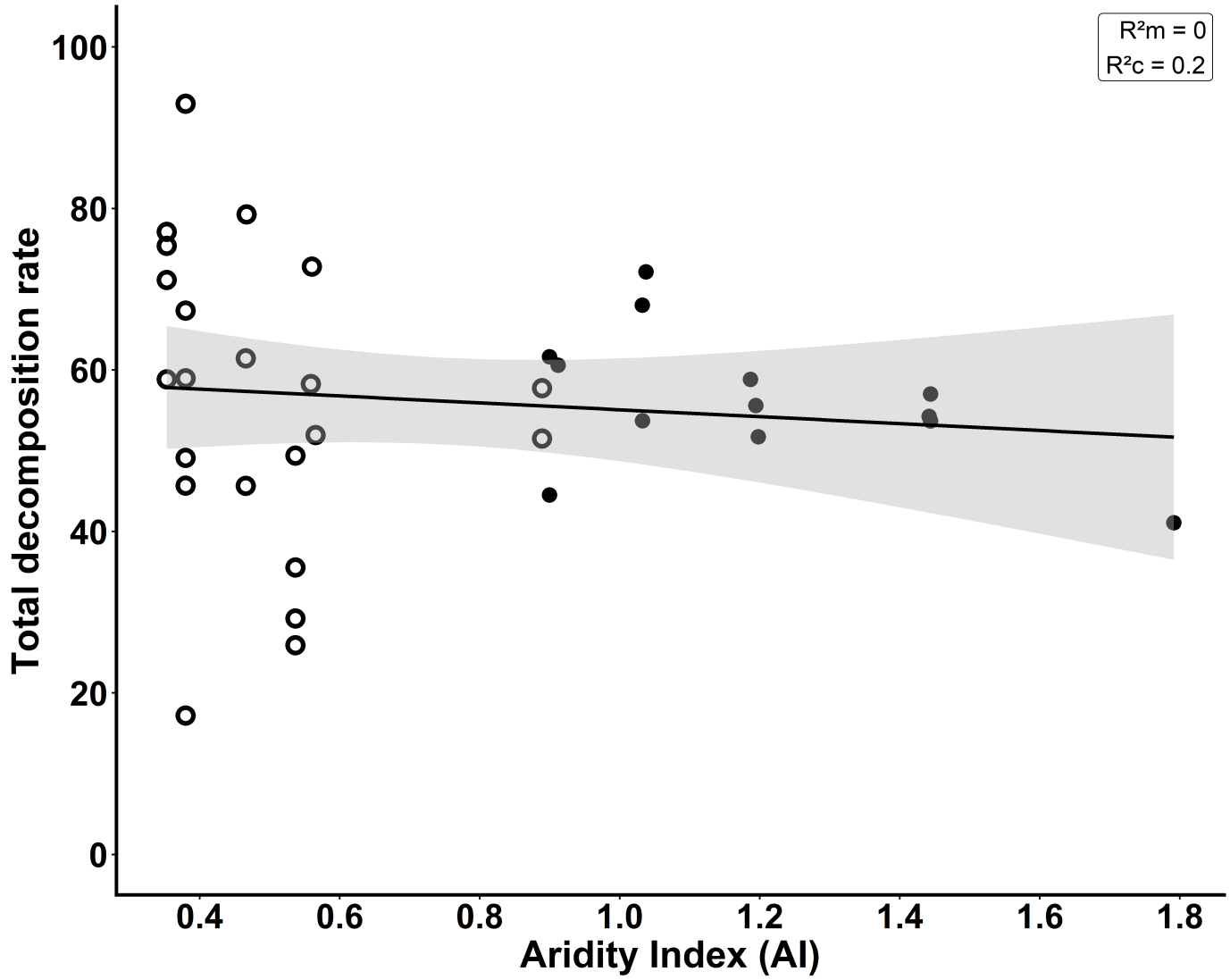
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
56.15	NA	NA	NA	NA	NA	3	-142.40	295.78	0.00	0.32
16.48	NA	1.59	NA	NA	NA	4	-140.49	296.95	1.17	0.18
6.96	-10.39	2.30	NA	NA	NA	5	-136.68	297.95	2.17	0.11
60.52	-5.53	NA	NA	NA	NA	4	-139.16	297.96	2.18	0.11
55.99	NA	NA	+	NA	NA	4	-139.63	298.41	2.63	0.09
22.31	-21.23	2.28	+	NA	NA	6	-132.96	299.49	3.71	0.05
9.75	NA	1.80	+	NA	NA	5	-137.59	299.58	3.79	0.05
73.97	-15.30	NA	+	NA	NA	5	-135.49	299.70	3.92	0.05
0.97	-13.08	2.72	+	+	NA	7	-128.23	301.57	5.78	0.02
-104.67	NA	6.21	+	NA	+	6	-134.11	302.56	6.78	0.01
93.21	-21.87	-0.43	+	NA	+	7	-129.52	302.78	7.00	0.01
72.85	-14.49	NA	+	+	NA	6	-131.03	302.83	7.05	0.01
40.09	-13.52	1.22	+	+	+	8	-124.81	305.42	9.64	0.00

Selected_model_formula
Term_Loss ~ (1 | Site)

nobs	AIC	BIC	logLik	sigma
35	290.79	295.46	-142.4	13.94

term	estimate	std_error	df	t_value
(Intercept)	56.15	3.31	8.75	16.98

R2m	R2c
0	0.2



4 4) Termite proportional (Term_Prop)

(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
89.04	-13.80	NA	+	+	NA	6	-114.47	265.48	0.00	0.30
95.87	-19.92	NA	NA	NA	NA	4	-123.76	265.89	0.41	0.25
108.45	-28.99	NA	+	NA	NA	5	-120.02	266.67	1.18	0.17
121.65	-17.84	-1.10	NA	NA	NA	5	-122.24	267.68	2.19	0.10
134.88	-26.96	-1.12	+	NA	NA	6	-118.49	268.53	3.04	0.07
104.33	-14.55	-0.56	+	+	NA	7	-113.37	268.61	3.13	0.06
115.37	-26.54	-0.39	+	NA	+	7	-115.33	271.92	6.44	0.01
18.82	-12.59	2.65	+	+	+	8	-110.23	272.30	6.81	0.01

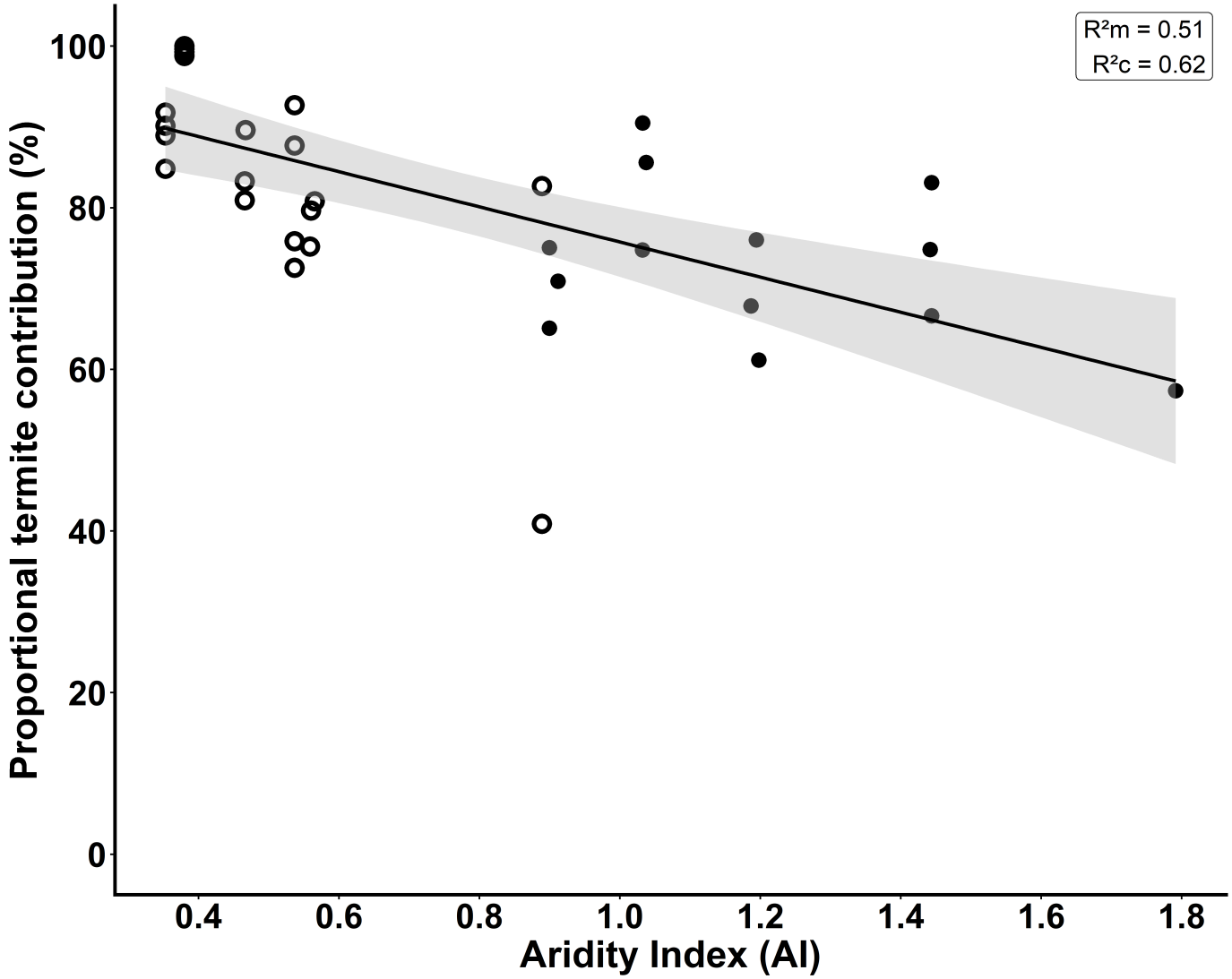
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
79.20	NA	NA	NA	NA	NA	3	-130.57	272.35	6.87	0.01
134.53	NA	-2.21	NA	NA	NA	4	-128.20	272.63	7.15	0.01
76.53	NA	NA	+	NA	NA	4	-127.68	274.05	8.57	0.00
127.37	NA	-2.00	+	NA	NA	5	-125.44	274.72	9.23	0.00
-212.08	NA	11.11	+	NA	+	6	-121.35	275.95	10.47	0.00

Selected_model_formula
Term_Prop ~ AI + VegType + (1 | Site) + AI:VegType

nobs	AIC	BIC	logLik	sigma
35	240.94	250.27	-114.47	8.32

term	estimate	std_error	df	t_value
(Intercept)	89.04	14.01	11.98	6.36
AI	-13.80	11.27	13.42	-1.23
VegTypeSavanna	24.29	17.63	7.59	1.38
AI:VegTypeSavanna	-42.59	20.95	8.30	-2.03

R2m	R2c
0.51	0.62



5 5) Microbe proportional (Microbe_Prop)

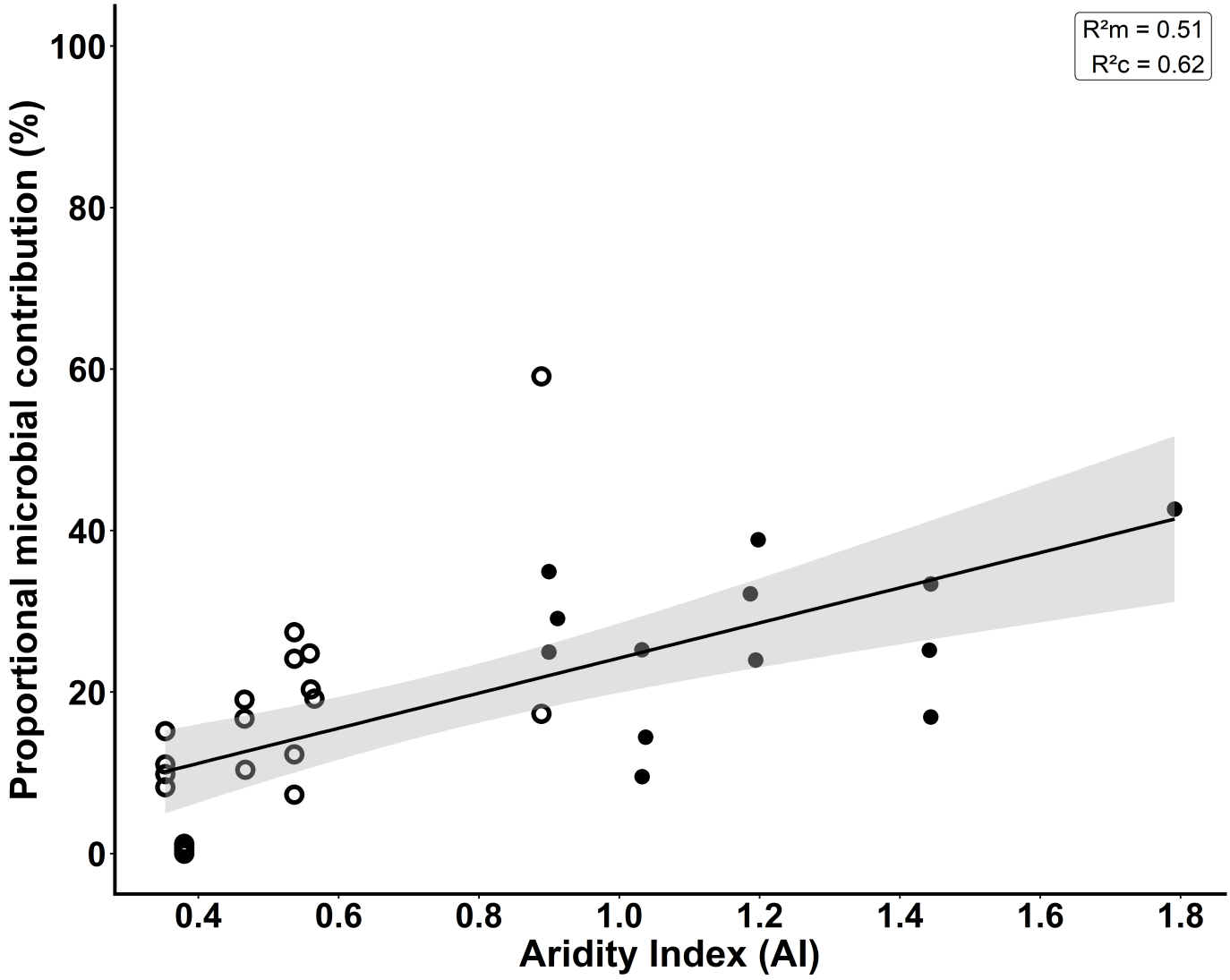
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
10.96	13.80	NA	+	+	NA	6	-114.47	265.48	0.00	0.30
4.13	19.92	NA	NA	NA	NA	4	-123.76	265.89	0.41	0.25
-8.45	28.99	NA	+	NA	NA	5	-120.02	266.67	1.18	0.17
-21.65	17.84	1.10	NA	NA	NA	5	-122.24	267.68	2.19	0.10
-34.88	26.96	1.12	+	NA	NA	6	-118.49	268.53	3.04	0.07
-4.33	14.55	0.56	+	+	NA	7	-113.37	268.61	3.13	0.06
-15.37	26.54	0.39	+	NA	+	7	-115.33	271.92	6.44	0.01
81.18	12.59	-2.65	+	+	+	8	-110.23	272.30	6.81	0.01
20.80	NA	NA	NA	NA	NA	3	-130.57	272.35	6.87	0.01
-34.53	NA	2.21	NA	NA	NA	4	-128.20	272.63	7.15	0.01
23.47	NA	NA	+	NA	NA	4	-127.68	274.05	8.57	0.00
-27.37	NA	2.00	+	NA	NA	5	-125.44	274.72	9.23	0.00
312.08	NA	-11.11	+	NA	+	6	-121.35	275.95	10.47	0.00

Selected_model_formula
Microbe_Prop ~ AI + VegType + (1 | Site) + AI:VegType

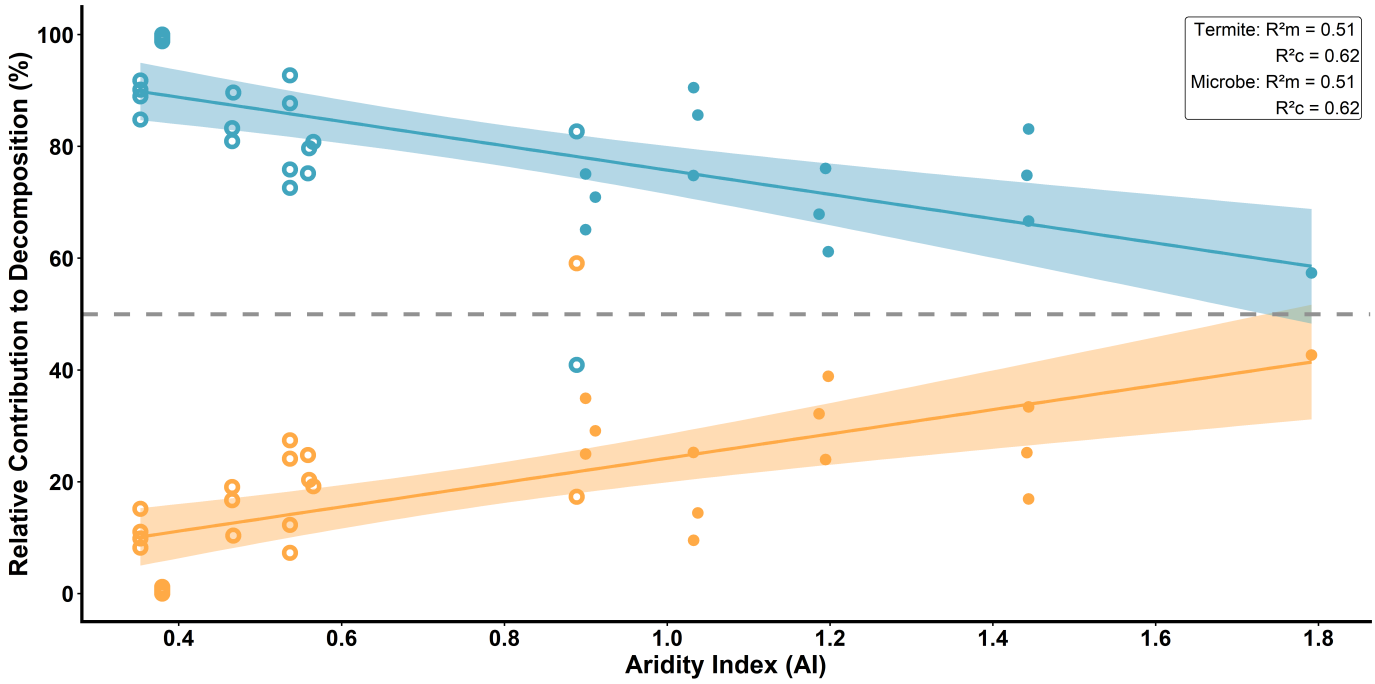
nobs	AIC	BIC	logLik	sigma
35	240.94	250.27	-114.47	8.32

term	estimate	std_error	df	t_value
(Intercept)	10.96	14.01	11.98	0.78
AI	13.80	11.27	13.42	1.23
VegTypeSavanna	-24.29	17.63	7.59	-1.38
AI:VegTypeSavanna	42.59	20.95	8.30	2.03

R2m	R2c
0.51	0.62

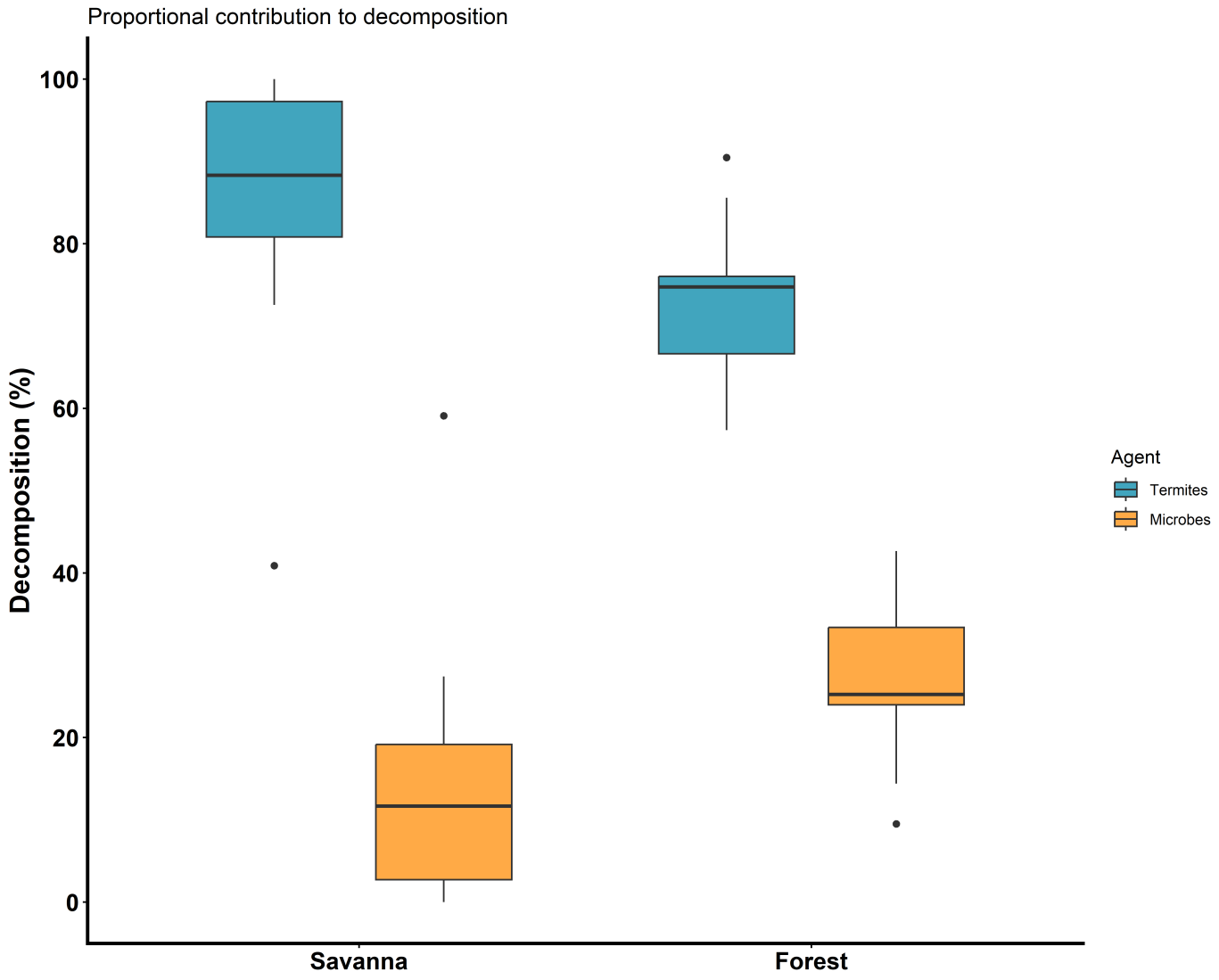


6 6) Combined proportional plot (termite + microbe)

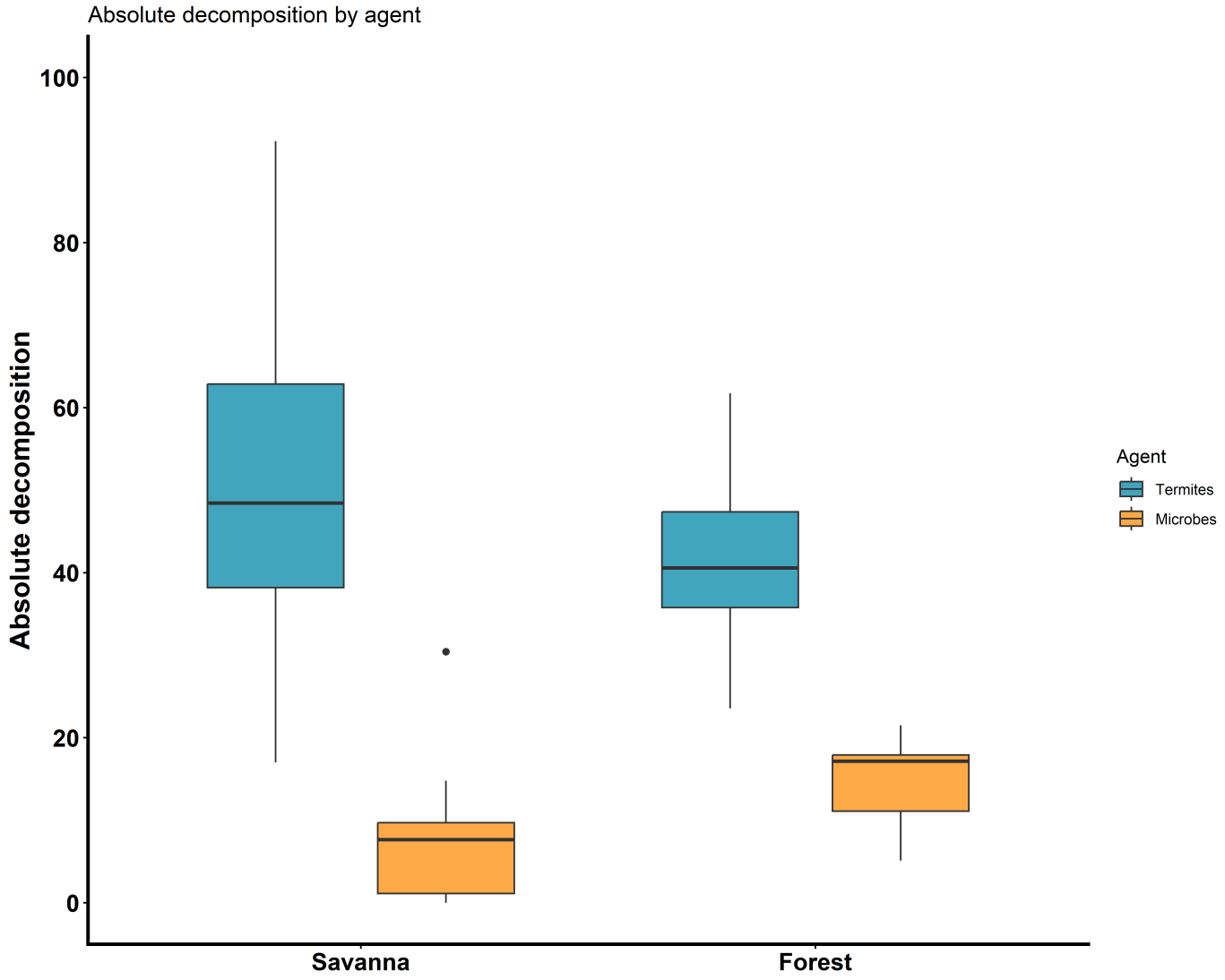


7 7) Boxplots (no points; filled by agent)

7.1 Proportional contributions (0–100)



7.2 Absolute decomposition (0–100)



7.3 Two-way ANOVA: absolute decomposition by

habitat type and decomposition agent

```
library(dplyr)
library(tidyr)
library(car)
library(knitr)

Final_DB <- Final_DB %>%
  mutate(
    VegType = as.factor(VegType)
  )

abs_decomp_long <- Final_DB %>%
  select(VegType, Term_Absolute, Microbe_Absolute) %>%
  pivot_longer(
    cols = c(Term_Absolute, Microbe_Absolute),
    names_to = "Decomposition_type",
    values_to = "Absolute_decomposition"
  ) %>%
  mutate(
    Decomposition_type = factor(
      Decomposition_type,
      levels = c("Term_Absolute", "Microbe_Absolute"),
      labels = c("Termite", "Microbial")
    )
  ) %>%
  filter(
    !is.na(VegType),
    !is.na(Decomposition_type),
    !is.na(Absolute_decomposition)
  )

abs_decomp_aov <- aov(
  Absolute_decomposition ~ VegType * Decomposition_type,
  data = abs_decomp_long
)

abs_decomp_anova <- as.data.frame(car::Anova(abs_decomp_aov, type = 2)) %>%
  tibble::rownames_to_column("term")

knitr::kable(
  abs_decomp_anova,
  digits = 4,
  caption = "Two-way ANOVA for absolute decomposition by habitat type and decomposition agen
t"
)
```

Two-way ANOVA for absolute decomposition by habitat type and decomposition agent

term	Sum Sq	Df	F value	Pr(>F)
VegType	0.4679	1	0.0030	0.9563
Decomposition_type	22355.3700	1	144.5079	0.0000
VegType:Decomposition_type	878.8965	1	5.6813	0.0200

term	Sum Sq	Df	F value	Pr(>F)
Residuals	10210.1957	66	NA	NA

7.4 Mean absolute decomposition by habitat type

```
abs_decomp_summary <- abs_decomp_long %>%
  group_by(VegType, Decomposition_type) %>%
  summarise(
    n = n(),
    mean = mean(Absolute_decomposition, na.rm = TRUE),
    sd = sd(Absolute_decomposition, na.rm = TRUE),
    se = sd / sqrt(n),
    .groups = "drop"
  )

knitr::kable(
  abs_decomp_summary,
  digits = 3,
  caption = "Mean absolute decomposition by habitat type and decomposition agent"
)
```

Mean absolute decomposition by habitat type and decomposition agent

VegType	Decomposition_type	n	mean	sd	se
Forest	Termite	13	41.443	9.610	2.665
Forest	Microbial	13	14.921	4.809	1.334
Savanna	Termite	22	48.607	19.306	4.116
Savanna	Microbial	22	7.418	6.892	1.469

7.5 Overall mean absolute decomposition by decomposition agent

```
overall_abs_decomp_summary <- abs_decomp_long %>%
  group_by(Decomposition_type) %>%
  summarise(
    n = n(),
    mean = mean(Absolute_decomposition, na.rm = TRUE),
    sd = sd(Absolute_decomposition, na.rm = TRUE),
    se = sd / sqrt(n),
    .groups = "drop"
  )

knitr::kable(
  overall_abs_decomp_summary,
  digits = 3,
  caption = "Overall mean absolute decomposition for termite and microbial decomposition"
)
```

Overall mean absolute decomposition for termite and microbial decomposition

Decomposition_type	n	mean	sd	se
Termite	35	45.946	16.587	2.804
Microbial	35	10.205	7.144	1.207